Tezpur University
Spring Semester End examination-2024.

Course Code: CO216 **Course Title: Formal Language and Automata**
Full marks=60 *Time = 3 hours*

(The figures in the right-hand margin indicate full marks for the questions)

****All questions are compulsory**

Q1.	Give deterministic finite automata (DFA) accepting the following languages over the alphabet {0,1} ***(Answer any 2):*** a) The set of all strings where every 0 is immediately preceded and followed by 1. b) The set of all strings that do not end in 01. c) The set of all strings containing either even number of 0's or odd number of 1's but not both. d) The set of all strings such that the 4th symbol from the right end is 1.	5 X 2 = 10
Q2.	Construct context-free grammar (CFG) that generate the following languages ***(Answer any 3):*** a) $\{w \in \{0,1\}^* \mid w = w^R \text{ and } \lvert w \rvert \text{ is even}\}$ b) $\{w \in \{0,1\}^* \mid$ the length of w is odd, and the middle symbol is 0$\}$ c) $\{w \in \{0,1\}^* \mid$ w contains at least three 1's$\}$ d) $\{a^i\, b^j\, c^k \mid i, j, k \geq 0 \text{ and } i + j = k\}$	5 X 3 = 15
Q3.	State True or False: a) If L1 and L2 are context free languages, then L1 U L2 is also context free. b) Context sensitive languages are closed under Kleene Closure. c) A Context Free Grammar is ambiguous if it has more than one leftmost derivation. d) Stack is an example of application of Finite State Machine.	1 x 4 = 4
Q4.	Prove that the language $L = \{w\, w^R\, w \mid w\, \varepsilon\{0,1\}^*\}$ is not context free using pumping lemma.	6
Q5.	Construct PDA for the given CFG, and test whether the string '010000' is acceptable by this PDA. $S \rightarrow 0BB$ $B \rightarrow 0S \mid 1S \mid 0$	4 +3 = 7
Q6.	Consider the grammar G: $S \rightarrow AB \mid BC$ $A \rightarrow BA \mid a$ $B \rightarrow CC \mid b$ $C \rightarrow AB \mid a$ Check if the string ***'baaba'*** is in L(G) using CYK algorithm.	6

Q7.	Define a Turing Machine (TM). Construct a TM for the language $L = \{a^n b^m a^{(n+m)} \mid n, m \geq 1\}$	6
Q8.	Construct an equivalent FA for the given regular expression (0+1)* (00+11) (0+1)*	4
Q9.	Explain decidability of a Turing Machine with respect to Halting problem.	2